

Exercise Animation System — Final Evolution Lab

Phase 3 Deliverable | Complete exercise & workout system with 23 animated exercises, 5 workout programs, and progressive overload logic.

Overview

The Exercise Animation System provides a complete training framework within Final Evolution Lab, combining 3D-animated exercise demonstrations, structured workout programs, and progress tracking.

Key Features

- **23 fully animated exercises** across 5 categories
- **5 pre-built workout programs** (Basketball Performance, Strength Foundation, Athletic Conditioning, Mobility & Recovery, Dunk Training)
- **Custom workout builder** — create and save personal programs
- **Progressive overload** — linear, undulating, and block periodization
- **Personal records** — track max reps, weight, and best times
- **Calorie tracking** — per-exercise and per-session estimates
- **Form validation** — real-time checkpoints during demonstrations
- **Integrated with all 17 game modes** — sport-specific warm-ups and training

Exercise Categories (23 Total)

Warm-Up (4 exercises)

ID	Name	Target Muscles	Duration
dynamic_lunge_walk	Dynamic Lunge Walk	Hip flexors, glutes	60s
arm_circles_progressive	Progressive Arm Circles	Deltoids, rotator cuff	45s
high_knees	High Knees	Hip flexors, quads	30s
lateral_shuffle	Lateral Shuffle	Adductors, glutes	30s

Strength Training (4 exercises)

ID	Name	Target Muscles	Sets x Reps
squat_form	Perfect Form Squat	Quads, glutes, core	4x8
box_jump	Box Jump	Quads, calves, glutes	4x6
pushup_explosive	Explosive Push-Up	Chest, triceps, core	4x10
deadlift_hip_hinge	Hip Hinge / Deadlift	Hamstrings, glutes, back	3x8

Mobility (4 exercises)

ID	Name	Target Area	Duration
hip_90_90	90/90 Hip Rotation	Hips	45s/side
thoracic_rotation	Thoracic Spine Rotation	Upper back	30s/side
ankle_dorsiflexion	Ankle Dorsiflexion Drill	Ankles	30s/side
shoulder_dislocate	Shoulder Pass-Through	Shoulders	30s

Sport-Specific (8 exercises)

ID	Name	Sport	Difficulty
dunk_approach	Dunk Approach Pattern	Basketball	0.7
golf_swing_drill	Golf Swing Plane Drill	Golf	0.6
kick_form	Soccer Kick Form	Soccer	0.5
serve_motion	Tennis Serve Motion	Tennis	0.6
karate_kata_basic	Basic Kata Sequence	Karate	0.5
surf_popup	Surf Pop-Up Drill	Surfing	0.4
balance_board	Balance Board Training	All	0.3
tumbling_roundoff	Gymnastics Roundoff	Gymnastics	0.7

Recovery / Cooldown (3 exercises)

ID	Name	Focus	Duration
foam_roll_it_band	IT Band Foam Roll	Legs	120s/side
pigeon_stretch	Pigeon Stretch	Hips	90s/side
breathing_box	Box Breathing	Nervous system	5 min

Workout Programs

1. Basketball Performance (8 weeks)

- **Goal:** Vertical jump, speed, agility
- **Days/week:** 4
- **Level:** Intermediate
- **Periodization:** Daily Undulating
- **Includes:** Deload week at week 4

2. Strength Foundation (12 weeks)

- **Goal:** Build base strength
- **Days/week:** 3
- **Level:** Beginner
- **Periodization:** Linear
- **Includes:** Deload week at week 4, 8, 12

3. Athletic Conditioning (6 weeks)

- **Goal:** Endurance and conditioning
- **Days/week:** 5
- **Level:** Advanced
- **Periodization:** Daily Undulating
- **Includes:** HIIT circuits, sport drills

4. Mobility & Recovery (4 weeks)

- **Goal:** Flexibility, injury prevention
- **Days/week:** 7 (daily)
- **Level:** Beginner
- **Includes:** Morning flow + evening recovery

5. Dunk Training (10 weeks)

- **Goal:** Vertical jump maximization
- **Days/week:** 4
- **Level:** Intermediate
- **Periodization:** Block
- **Includes:** Max strength, plyometrics, technique

Architecture

C++ Classes

File	Class	Purpose
FELExercise.h	FExerciseData , enums	Core data structures
FELExerciseManager.h/.cpp	UFELExerciseManager	Database, animation loading, progress
FELWorkoutProgram.h/.cpp	UFELWorkoutProgramManager	Workout programs, sessions, overload

Data Files

File	Content
Content/FEL/Config/ExerciseCatalog.json	23 exercise definitions
Content/FEL/Data/WorkoutCatalogues.json	5 pre-built workout programs
Saved/FEL/exercise_progress.json	Player progress (auto-saved)
Saved/FEL/custom_programs.json	User-created programs

UI Widgets

Widget	Purpose
FELExerciseLibraryWidget	Browse all 23 exercises with filtering
FELWorkoutCatalogueWidget	Browse and select workout programs
FELWorkoutSessionWidget	Active workout session tracking

Progressive Overload

The system supports three periodization schemes:

Linear

- Adds 2.5–5% weight each week
- Falls back to +1 rep if no weight is used

Daily Undulating (DUP)

- Alternates heavy (5 reps) → light (12 reps) → moderate (8 reps)

- Adjusts weight inversely to rep count

Block

- Adds volume (sets) rather than weight
- Caps at 6 sets per exercise

Game Mode Integration

Each of the 17 game modes maps to relevant exercises via `sportRelevance` tags. Before starting a game mode, the system can suggest a warm-up routine based on the sport.

```
// Get exercises relevant to basketball
TArray<FExerciseData> BasketballExercises =
    ExerciseManager->GetExercisesForGameMode(TEXT("basketball_h2h"));
```

API Reference

UFELExerciseManager

```
GetAllExercises()           // Returns all 23 exercises
GetExercisesByCategory(Category) // Filter by category
GetExercisesByDifficulty(Level) // Filter by difficulty
GetExerciseByID(ID, OutExercise) // Lookup by ID
GetExercisesForGameMode(ModeID) // Sport-specific exercises
SearchExercises(Query)      // Text search
LoadExerciseAnimation(ID, Variant) // Load animation montage
RecordExerciseCompletion(...) // Log completion + PR check
GetExerciseProgress(ID)     // Get progress stats
CalculateCaloriesBurned(ID, Mins) // Calorie estimate
GetRecommendedExercises(Count) // Smart recommendations
```

UFELWorkoutProgramManager

```
GetAllPrograms()           // Returns all 5 programs
GetProgramByID(ID, OutProgram) // Lookup by ID
GetProgramsByGoal(Goal)    // Filter by goal
StartWorkoutSession(ID, Week, Day) // Begin workout
CompleteCurrentExercise(Reps, Wt) // Complete & advance
EndWorkoutSession()        // Finish session
CalculateProgressiveOverload(...) // Next week's prescription
CreateCustomProgram(Name, Goal, Days) // Custom builder
```